SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.2: Animal Transport — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 3.2: Animal Transport
Driving Question	DRIVING QUESTION: How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon: Eliud Kipchoge and the Limits of the Body**	Students observed a video clip or image set of Eliud Kipchoge completing a marathon and were introduced to the anchoring phenomenon: his heart beats ~180 times per minute during peak exertion, pumping oxygenated blood to every working muscle with extreme efficiency. Students also observed a grasshopper moving and were told it has no blood vessels — haemolymph simply floods its body cavity. The contrast between these two animals anchored the driving question.	Students learned that different animals have fundamentally different transport systems, and that the type of system an animal has is directly linked to its size, metabolic rate, and lifestyle. They learned that Eliud Kipchoge's extraordinary endurance depends on a closed, double circulatory system that can sustain high oxygen delivery — something a grasshopper's open system cannot do. This lesson established the need-to-know questions that will drive the unit.	The teacher used this lesson to surface prior knowledge and elicit students' initial mental models of how the body 'moves things around.' Pedagogical note: Expect students to conflate the heart with the lungs, or to think blood simply 'flows' without a pump. Record these misconceptions on the Driving Question Board (DQB) without correcting them yet — they will be resolved across subsequent lessons. Cold-call at least four students to explain what they think happens inside Kipchoge's chest during a race. This lesson should end with more questions than answers.
Lesson 2 **Why Can't Large Animals Just Use Diffusion? — The Importance of Transport Systems**	Students observed a demonstration or simulation comparing surface-area-to-volume (SA:V) ratios using cubes of different sizes (e.g., 1 cm^3, 2 cm^3, 4 cm^3 agar blocks stained with potassium permanganate). They measured how far the stain diffused into each block within a fixed time and recorded that diffusion distance remained constant while the volume grew — meaning the interior of larger blocks remained unstained. They also examined	Students learned that as an organism's size increases, its SA:V ratio decreases, making passive diffusion insufficient to supply all cells with oxygen and nutrients or remove waste. They learned that transport systems evolved to solve this problem by providing bulk flow — moving substances rapidly over long distances regardless of body size. They also learned that unicellular organisms like amoeba rely entirely on diffusion, which works only because the	The teacher should connect this directly to Kipchoge: his body contains ~37 trillion cells, none of which are close enough to a body surface to receive oxygen by diffusion alone. Ask students: 'If Kipchoge were the size of an amoeba, would he need a heart?' (No — diffusion would suffice.) Pedagogical note: The SA:V concept is mathematically challenging; allocate extra wait time (at least 10 seconds) after posing ratio questions. Use the maize grain vs. maize

	data comparing single-celled organisms (amoeba) with multicellular animals (a cow, a human, a tilapia from Lake Victoria).	distances involved are tiny.	sack analogy — a single maize grain absorbs water quickly, but a tightly packed sack does not penetrate evenly — to make the concept locally concrete.
Lesson 3 **Transport in Insects: The Open Circulatory System of the Grasshopper**	Students observed labelled diagrams and/or a video of a grasshopper's open circulatory system, identifying the dorsal vessel (heart), ostia, and haemocoel. They noted that the haemolymph does not travel in a closed circuit of vessels but instead floods the body cavity, bathing organs directly. They also observed that insects deliver oxygen NOT through haemolymph but through a separate tracheal system of air tubes — a key distinction from vertebrates.	Students learned that an open circulatory system moves haemolymph at low pressure through an open body cavity (haemocoel), and that this system is adequate for small, low-metabolic-rate animals like the grasshopper. They learned that because oxygen delivery is handled by the tracheal system, the grasshopper's circulatory system is primarily responsible for transporting nutrients, hormones, and waste — not oxygen. This explains why the grasshopper does not need high-pressure, high-speed circulation.	The teacher should use this lesson to answer a key DQB question: 'Why can't Kipchoge have a grasshopper's heart?' Students should articulate that a grasshopper's open system cannot generate the high blood pressure needed to deliver oxygen at the rate Kipchoge's muscles demand during a 42 km race. Pedagogical note: Students frequently confuse haemolymph with blood — clarify that haemolymph lacks haemoglobin and does not carry oxygen in insects. Have students revise their initial models on the DQB, crossing out misconceptions and adding the open/closed distinction.
Lesson 4 **Transport in Fish — The Single Circulatory System of the Tilapia**	Students observed a dissection (or detailed diagram/video) of a tilapia (Oreochromis niloticus) from Lake Victoria, identifying its two-chambered heart (one atrium, one ventricle), gills, and the single loop of blood flow: heart → gills (oxygenation) → body tissues → heart. They traced the path of a red blood cell on a diagram and noted that blood passes through the heart only once per complete circuit.	Students learned that the tilapia has a closed but single circulatory system. Blood is oxygenated in the gills and then travels directly to body tissues before returning to the heart — meaning it loses pressure after passing through the gill capillaries, arriving at body tissues at relatively low pressure. They learned that this system is more efficient than an open system but less efficient than a double circulatory system because blood is not re-pressurised before reaching the body.	The teacher should prompt students to compare the tilapia's system with the grasshopper's: both are limited in how much oxygen they can deliver per unit time, but for different reasons. Connect to the driving question: Kipchoge runs at ~21 km/h for over two hours — a tilapia's single circuit could not sustain this because muscles would receive low-pressure, partially deoxygenated blood. Pedagogical note: Use a Nairobi water distribution analogy — a pump that pushes water through two substations loses pressure at each stage; a double-pump system re-pressurises at each stage.
Lesson 5 **Transport in Amphibians and Reptiles — The Trade-Off Heart**	Students observed diagrams of the frog (amphibian) and crocodile (reptile) hearts, comparing the three-chambered frog heart (two atria, one ventricle — allowing mixing of oxygenated and deoxygenated blood) with the almost-four-chambered crocodile heart (nearly complete ventricular septum). They traced blood flow in each, noting where mixing occurs and where separation is maintained.	Students learned that the frog's three-chambered heart represents an evolutionary trade-off: it supports a more active lifestyle than fish, but the mixing of oxygenated and deoxygenated blood in the single ventricle reduces the efficiency of oxygen delivery. They learned that reptiles (e.g., Nile crocodile) show a more complete ventricular septum, reducing mixing and increasing efficiency — reflecting greater metabolic demands. They understood this	The teacher should frame this lesson as the 'missing link' in the evolutionary story of circulatory systems. Ask: 'Why does the frog get away with mixing blood in one ventricle?' (It is ectothermic — low metabolic demand; also absorbs oxygen through its skin.) Cold-call at least three students to rank the five systems studied so far (amoeba/diffusion, grasshopper, tilapia, frog, crocodile) in order of oxygen-delivery efficiency and justify their ranking.

		as an evolutionary continuum from open systems (insects) toward the fully separated four-chambered mammalian heart.	Pedagogical note: Students often think more chambers always means better — help them see that the frog's system is appropriate FOR the frog's lifestyle.
Lesson 6 **Transport in Mammals — The Four-Chambered Heart and Double Circulatory System**	Students observed a dissection (or high-quality diagram/video) of a mammalian heart (sheep or cattle heart, commonly available in Kenyan markets), identifying the four chambers (right atrium, right ventricle, left atrium, left ventricle), valves (tricuspid, bicuspid/mitral, semilunar), coronary arteries, vena cava, pulmonary artery and vein, and aorta. They traced the pulmonary circuit (heart → lungs → heart) and systemic circuit (heart → body → heart) separately.	Students learned that the mammalian double circulatory system maintains two separate circuits, each re-pressurised by a dedicated ventricle. The right ventricle pumps deoxygenated blood to the lungs at low pressure; the left ventricle pumps oxygenated blood to the body at high pressure. This complete separation eliminates blood mixing and enables sustained high oxygen delivery — the physiological foundation of endurance performance. Students also learned that the left ventricular wall is thicker than the right, reflecting the greater force needed to push blood through the systemic circuit.	The teacher should make explicit the connection to Eliud Kipchoge: his left ventricle is enlarged through years of training (cardiac hypertrophy), enabling it to pump more blood per beat (high stroke volume). Ask: 'Now that we know the four-chambered heart, can you explain in one sentence why Kipchoge cannot survive a marathon with a grasshopper's circulatory system?' Students should be able to produce a complete explanation. Pedagogical note: This is a landmark lesson — ensure the DQB is updated with the open/closed, single/double distinctions clearly posted. This lesson anchors all subsequent lessons on heart mechanics.
Lesson 7 **Pumping Mechanism of the Mammalian Heart: The Cardiac Cycle, Heart Sounds, and the SAN**	Students observed an animation or diagram of the cardiac cycle showing three phases: atrial systole (atria contract, pushing blood into ventricles), ventricular systole (ventricles contract, ejecting blood into arteries), and diastole (all chambers relax, filling with blood). They listened to or observed a recording of 'lub-dub' heart sounds and identified that 'lub' is caused by atrioventricular valve closure and 'dub' by semilunar valve closure. They also traced the electrical conduction pathway: SAN (sinoatrial node — the pacemaker) → AVN (atrioventricular node) → Bundle of His → Purkinje fibres.	Students learned that the heart is myogenic — it generates its own electrical impulse at the SAN without requiring a signal from the brain. They learned that the cardiac cycle is a precisely timed sequence ensuring atria contract first (filling ventricles) and ventricles contract second (ejecting blood). They learned that heart sounds are diagnostic tools: a murmur (abnormal sound) may indicate a faulty valve. Students connected cardiac output (CO = stroke volume × heart rate) to Kipchoge's performance: at peak exertion, his heart rate rises and stroke volume increases, maximising CO.	The teacher should use a drum or clap to simulate the cardiac cycle rhythm in class. Ask students: 'If the SAN is damaged, what happens to the heart?' (It may beat irregularly or stop — this is why pacemakers are implanted.) Cold-call four students to trace the path of an electrical impulse from SAN to Purkinje fibres. Pedagogical note: Students often think the brain controls each heartbeat — explicitly correct this by emphasising myogenic origin. Connect to the driving question: if Kipchoge had a grasshopper's dorsal vessel (which lacks an SAN), his heart could not increase its rate on demand during a race.
Lesson 8 **The Lymphatic System: The Hidden Highway of Animal Transport**	Students observed a diagram of the lymphatic system showing lymph capillaries, lymph nodes (e.g., in the armpit and groin — locations students can feel themselves), lymph vessels, the thoracic duct, and the points of return to the bloodstream at the subclavian veins. They noted that lymph flows in only one direction (toward the heart) and is moved by skeletal	Students learned that plasma leaks out of capillaries into interstitial spaces to form tissue fluid. Most is reabsorbed by osmosis, but the remainder drains into lymph capillaries, becoming lymph. The lymphatic system returns this fluid to the blood, preventing oedema (dangerous swelling). Lymph nodes filter lymph, trapping pathogens and activating immune	The teacher should connect lymph to a real Kenyan health context: lymphatic filariasis (elephantiasis), caused by a parasitic worm blocking lymph vessels, is present in coastal Kenya (Mombasa region) and causes gross oedema of the limbs. Ask: 'What would happen to Kipchoge's leg muscles if his lymph vessels were blocked during a race?' (Tissue fluid would

	muscle contractions and one-way valves — not by a pump. They observed that lymph is colourless/pale yellow and forms from excess tissue fluid.	responses. Students learned that the lymphatic system is essential to completing the transport story — it closes the loop between the cardiovascular system and the tissues.	accumulate, causing swelling, reduced muscle function, and potentially dangerous pressure.) Pedagogical note: Students frequently omit the lymphatic system when describing transport — emphasise that it is a THIRD transport network alongside arteries/veins.
Lesson 9 **The Immune System and Blood Clotting: The Body's Defence and Repair Network**	Students observed microscope slides or diagrams of blood smears identifying erythrocytes (red blood cells — biconcave, no nucleus), leucocytes (white blood cells — phagocytes and lymphocytes), platelets, and plasma. They also observed a video or diagram of the clotting cascade: damaged tissue → platelets aggregate → thromboplastin released → prothrombin → thrombin → fibrinogen → fibrin mesh → clot. They examined data on how long clotting takes and what happens in haemophilia (clotting factor absent).	Students learned that blood performs three integrated functions: delivery (RBCs carry oxygen via haemoglobin and carry CO_2), defence (phagocytes engulf pathogens; lymphocytes produce antibodies), and repair (platelets and clotting factors seal wounds). They learned that haemoglobin's affinity for oxygen is pH-dependent — in active muscles (lower pH due to CO_2 production), haemoglobin releases oxygen more readily (Bohr effect). This directly connects to Kipchoge's muscles receiving more oxygen precisely when they need it most.	The teacher should prompt students to consider what Kipchoge's blood is doing during the final kilometre of a marathon: his muscles are producing CO_2 rapidly, lowering blood pH, which causes haemoglobin to release even more oxygen — a self-reinforcing delivery mechanism. Ask: 'If a grasshopper's haemolymph lacks haemoglobin, how does it cope?' (Oxygen is delivered by trachea directly — haemolymph is not the oxygen carrier.) Pedagogical note: The Bohr effect is a KCSE favourite — ensure students can explain it in both directions (active muscle vs. resting lung).
Lesson 10 **ABO and Rhesus Factor Blood Grouping**	Students observed a blood-typing simulation (or actual ABO blood-typing using antisera A, B, and anti-D on cards), recording agglutination results for four unknown samples and identifying blood groups as A, B, AB, or O, and Rh+ or Rh−. They observed a data table showing the distribution of blood groups in Kenya (approximately: O = 47%, A = 26%, B = 23%, AB = 4%) and noted which groups are universal donors and universal recipients.	Students learned that red blood cells carry surface antigens (A, B, both, or neither) and plasma carries corresponding antibodies (anti-A, anti-B). Transfusion of incompatible blood causes agglutination (clumping) and haemolysis — potentially fatal. They learned that group O (no antigens) is the universal donor and group AB (no antibodies) is the universal recipient. They also learned that the Rhesus factor (D antigen) is critical in pregnancy — an Rh− mother carrying an Rh+ baby can develop anti-D antibodies, causing haemolytic disease of the newborn. This is clinically managed at Kenyan hospitals like Kenyatta National Hospital.	The teacher should connect blood grouping to real Kenyan public health contexts: road accident victims at Nairobi's major hospitals urgently need type O blood; blood donation drives at schools and universities save lives. Ask students: 'If Kipchoge were in an accident and needed an emergency transfusion, why would knowing his blood group matter?' Pedagogical note: Students often confuse antigens (on cells) with antibodies (in plasma) — use a 'lock and key' visual: antigen is the lock, antibody is the key. When the wrong key is inserted, the lock jams (agglutination).
Lesson 11 **Diseases of the Transport System and Public Health in Kenya**	Students observed case studies and data on transport system diseases prevalent in Kenya, including: hypertension (high blood pressure — leading cause of stroke in Kenya), anaemia (low RBC/haemoglobin count — widespread in children in Western Kenya), sickle cell disease (prevalent in	Students learned how structural or functional failures of the transport system cause disease: hypertension results from narrowed/stiff arteries increasing resistance; anaemia results from insufficient haemoglobin reducing oxygen delivery; sickle cell disease results from abnormal	The teacher should frame each disease as a 'malfunction of a structure we have already studied.' Ask: 'If Kipchoge had severe anaemia, how would it affect his marathon performance?' (Less haemoglobin → less oxygen per litre of blood → muscles fatigue faster → slower

	malaria-endemic regions including Kisumu and the Lake Victoria basin), and coronary heart disease (blocked coronary arteries — increasing with urbanisation in Nairobi). They also examined ECG traces to identify normal vs. abnormal heart rhythms.	haemoglobin causing RBCs to deform under low-oxygen conditions, blocking capillaries; coronary heart disease results from atherosclerotic plaques blocking coronary arteries, depriving heart muscle of oxygen. Students understood that each disease is a breakdown of a specific transport function studied in the unit.	pace or inability to finish.) Pedagogical note: Sickle cell disease has a protective heterozygous advantage against malaria — this is a powerful example connecting transport biology to genetics and ecology. Ensure students do not stigmatise sickle cell carriers; use the protective advantage framing positively.
Lesson 12 **Final Model Presentations & Celebratory Sensemaking: Answering the Driving Question**	Students presented their cumulative transport system models (built and revised across all 12 lessons) to the class, and the class collaboratively reviewed the Driving Question Board to confirm which questions had been answered. Students observed a final comparative diagram showing all five transport systems studied (diffusion/amoeba, open/grasshopper, single closed/tilapia, three-chambered/frog, double closed/mammal) on a single evolutionary continuum poster.	Students synthesised the following key understandings: (1) Transport systems vary from open (grasshopper) to closed-single (tilapia) to closed-double (mammal), with increasing efficiency of oxygen delivery. (2) The four-chambered mammalian heart, driven by the myogenic SAN, sustains the high cardiac output needed for endurance performance. (3) Blood performs delivery (haemoglobin), defence (leucocytes), and repair (platelets/clotting) functions. (4) The lymphatic system completes the transport network by draining tissue fluid and mounting immune responses. (5) Transport system diseases represent specific structural or functional failures of these systems. (6) Eliud Kipchoge's circulatory system — closed, double, four-chambered, with high stroke volume — is the direct physiological basis of his world-record marathon performance; a grasshopper's open system could not sustain this at any level.	The teacher should facilitate a whole-class 'answer the driving question' discussion in which students construct a collective, evidence-based response. The final answer should include: (a) structure of Kipchoge's system, (b) why a grasshopper's system is insufficient, and (c) what would actually happen biologically if Kipchoge had a grasshopper's circulatory system (haemolymph at low pressure, no haemoglobin, no re-pressurisation — he would collapse within metres of the start line). Pedagogical note: Celebrate student model revisions — display before-and-after models side by side. Cold-call at least six students to each contribute one sentence to the final driving question answer. End the unit by asking: 'What new questions does this raise?' to foster continued scientific curiosity.